

Course 2: Getting Started with Identity Governance and Administration (IGA)

Module 1: Introduction to IGA

What is Identity Governance and Administration (IGA)?

Identity Governance and Administration (IGA) is a cybersecurity process that manages digital identities and controls user access to applications, systems, and data within an organization.

IGA ensures that only authorized users can access the resources they need to perform their job, while preventing unauthorized access.

Simple Definition

IGA is a security framework that ensures the right user gets the right access to the right resource at the right time for the right reason.

Why Do We Need IGA?

Imagine a company with:

- 20,000 employees
- 500 applications
- Multiple cloud platforms
- Vendors
- Contractors

If IT manually creates accounts and grants access, it becomes difficult to:

- Track who has access
- Remove access when employees leave
- Meet audit requirements
- Prevent security risks

IGA solves these problems by automating identity and access management.

Objectives of IGA

IGA answers three important security questions.

1. Who has access?

Identifies which users have access to which applications.

Example:

Rahul has access to:

- Active Directory
- SAP
- Outlook

2. Why do they have access?

Ensures access is granted only for business purposes.

Example:

Rahul works in Finance.

Therefore, he requires SAP Finance access.

3. What are they doing with that access?

Monitors whether access is being used correctly.

Example:

Rahul can create invoices.

He cannot approve payments because that violates Segregation of Duties (SoD).

Main Functions of IGA

IGA performs several important tasks.

Identity Lifecycle Management

Manages users from the day they join until they leave.

This includes:

- Joiner
 - Mover
 - Leaver
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Access Management

Grants and removes application access automatically.

Access Reviews

Managers periodically verify whether users still require their access.

Access Certification

Confirms that existing access is still valid and compliant.

Segregation of Duties (SoD)

Detects conflicting permissions that may lead to fraud.

Compliance Management

Helps organizations comply with regulations such as:

- SOX

- HIPAA
 - GDPR
 - ISO 27001
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Importance of IGA

IGA mainly focuses on three areas.

Apply

Implements identity governance processes across all systems.

Example:

Automatically creating accounts for new employees.

Define and Enforce

Defines security policies and ensures they are followed.

Example:

Only HR Managers can approve employee termination.

Ensure

Ensures:

- Right User
- Right Access
- Right Resource
- Right Time

This is the core objective of every IGA solution.

What is an Identity?

An Identity is any person or entity that requires access to organizational resources.

Identities are of two types.

Human Identities

Examples:

- Employee
 - Student
 - Customer
 - Vendor
 - Contractor
 - Partner
-

Non-Human (Silicon) Identities

Examples:

- Service Accounts
- Bots
- APIs
- IoT Devices
- Robotic Process Automation (RPA)
- Applications

These identities also require authentication and authorization.

Example of Identity

Employee Rahul logs into Outlook.

Rahul is the Identity.

His Outlook account is the Account.

His Mailbox permission is the Entitlement.

Finance Role is the Role assigned to him.

Signs of a Successful IGA Program

A successful IGA implementation should:

- Automate user onboarding
 - Automatically remove unnecessary access
 - Reduce manual work
 - Support audits
 - Follow Least Privilege Principle
 - Reduce security risks
 - Improve user productivity
 - Lower operational cost
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Principle of Least Privilege (PoLP)

Definition

Users should receive only the minimum permissions required to perform their work.

Example

An HR employee needs:

- HR Portal
- Payroll System

They should NOT receive:

- Database Administrator
- Linux Root Access

This reduces attack surface and insider threats.

Why Saviynt is a Good IGA Platform

Saviynt provides modern identity governance using four major strengths.

Cloud Native

Designed specifically for cloud environments.

Benefits:

- High Availability
 - Scalability
 - Easy Upgrades
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Inherent Intelligence

Uses Artificial Intelligence and Machine Learning to:

- Recommend access
 - Detect risky permissions
 - Reduce manual effort
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Converged Platform

Instead of separate products for:

- IGA
- PAM
- ISPM
- AAG
- EIM

Saviynt combines everything into one platform.

Compliance Controls

Provides:

- Audit Reports
 - Access Reviews
 - Certification Campaigns
 - SoD Analysis
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Module 2: Saviynt Identity Cloud Architecture

What is Architecture?

Architecture describes how Saviynt is internally designed.

Saviynt follows a Cloud Native Microservices Architecture.

Instead of one large application, Saviynt is divided into multiple independent services.

Advantages of Microservices

Independent Scaling

Each service can be scaled separately.

Example:

Only Access Request Service receives heavy traffic.

Only that service is scaled.

Independent Deployment

New features can be deployed without affecting the whole application.

Better Performance

Multiple services run simultaneously.

Faster Upgrades

Small services are upgraded individually.

Better Reliability

If one service fails, others continue working.

Three Layers of Saviynt

1. User Interface Layer

Visible to end users.

Examples:

Dashboard

Reports

Access Requests

Search

2. Business Logic Layer

Processes requests.

Examples:

Approval Workflow

Provisioning Engine

Role Assignment

Policy Evaluation

3. Data Layer

Stores all identity data.

Examples:

Users

Accounts

Roles

Entitlements

Audit Logs

Identity Repository Schema

Definition

The Identity Repository Schema is Saviynt's database model.

It stores all identity-related information.

The schema defines:

- Tables
 - Relationships
 - Columns
 - Identity data
-

Important Database Tables

users

Stores user information.

Examples:

Employee ID

Department

Email

Manager

Status

accounts

Stores application accounts.

Example:

Active Directory Account

SAP Account

Salesforce Account

user_accounts

Connects Users with Accounts.

One User can own multiple Accounts.

account_entitlements1

Maps Accounts with Entitlements.

Example

SAP Account

↓

Finance Group

↓

Create Invoice Permission

entitlement_values

Stores all available permissions.

Examples

Admin

VPN

Finance

HR

Read Only

Module 3: Core IGA Concepts

IGA is built using four important objects.

User

Represents a person inside Saviynt.

Example

Rahul

Account

Represents a login inside a target application.

Examples

Rahul's AD Account

Rahul's SAP Account

Rahul's Salesforce Account

Entitlement

Represents a permission.

Examples

Read Access

Write Access

Finance Group

VPN Access

Role

A collection of multiple entitlements.

Example

Finance Role

Contains:

- SAP Login

- Finance Group
 - Report Access
 - Invoice Permission
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Relationship Between Objects

User



Account



Entitlements



Role

One Role contains multiple Entitlements.

One User can own multiple Accounts.

One Account can contain multiple Entitlements.

Module 4: Types of Applications

Saviynt supports three application types.

Connected Application

Uses connectors for:

- Reconciliation
- Provisioning

Everything is automatic.

Examples

Active Directory

Azure AD

AWS

SAP

Salesforce

Hybrid Application

Uses different connectors.

One connector imports data.

Another connector provisions access.

Disconnected Application

No connector exists.

Everything is manual.

Usually uses:

- CSV
 - Excel
 - SOAP
 - REST API
 - Manual Upload
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Provisioning vs Reconciliation

Provisioning

Direction

Saviynt → Target Application

Purpose

Creates or updates accounts.

Example

Employee joins.

Saviynt creates AD account.

Deprovisioning

Removes accounts or access.

Example

Employee resigns.

Saviynt disables SAP account.

Reconciliation

Direction

Target Application → Saviynt

Purpose

Imports users, accounts, and entitlements into Saviynt.

Example

Nightly Active Directory import.

Module 5: Saviynt Login

Two login methods exist.

Without SSO

User logs in using:

Username

Password

Authentication is performed directly by Saviynt.

With SSO

User is redirected to an Identity Provider.

Examples

Okta

Azure AD

PingFederate

After successful authentication, the user is redirected back to Saviynt.

Module 6: Saviynt Interface

The homepage contains:

- Dashboard
- Search Bar
- Navigation Menu
- Access Requests
- Applications
- Users

- Reports
- Settings
- Certifications

Dashboard tiles can be customized according to user preference.

Real-Time End-to-End Example

A new employee named Rahul joins the Finance department.

1. HR creates Rahul in Workday.
2. Saviynt imports Rahul as a User.
3. Saviynt provisions an Active Directory account.
4. SAP account is created.
5. Finance Role is assigned.
6. Finance Role grants multiple entitlements.
7. Every month, reconciliation verifies the accounts.
8. During certification, the manager reviews Rahul's access.
9. When Rahul leaves, Saviynt deprovisions all accounts automatically.

This entire process is called the **Identity Lifecycle**.

Quick Revision

- IGA manages identities and access.
- Identity can be human or non-human.
- User represents a person.
- Account is the login in an application.
- Entitlement is a permission.
- Role is a collection of entitlements.
- Provisioning creates or updates access.
- Reconciliation imports data into Saviynt.
- Deprovisioning removes access.
- Saviynt uses a cloud-native microservices architecture.
- Least Privilege means giving only the minimum required access.
- Connected applications use automated provisioning and reconciliation.
- Hybrid applications use separate connectors.
- Disconnected applications require manual integration.

Interview Questions

Q1. What is IGA?

IGA (Identity Governance and Administration) manages user identities, access, and permissions while ensuring security and compliance.

Q2. What are the four access building blocks in Saviynt?

- User
 - Account
 - Entitlement
 - Role
-

Q3. What is the difference between a User and an Account?

A **User** is the person's identity in Saviynt, while an **Account** is the login created for that user in a target application (such as Active Directory or SAP).

Q4. What is Provisioning?

Provisioning is the process of creating, updating, or assigning user accounts and access in target applications.

Q5. What is Reconciliation?

Reconciliation is the process of importing accounts and entitlements from target systems into Saviynt to ensure data is accurate and up to date.

Q6. What are the three types of applications in Saviynt?

- Connected Applications
- Hybrid Applications
- Disconnected Applications

Q7. What is the Principle of Least Privilege?

It is the security principle of giving users only the minimum access required to perform their job responsibilities.